Checkpoint #3: Josh

Role E. Fines, Notifications & Reports · COSC 3380 · Group 6 · Due Tuesday, October 27, 2026

Your schedule

Week	What you do
Week 1 (Oct 8 to 14)	Azure resource group, MySQL server, load the dump, create the web app, GitHub repo and skeleton. Write `sample_data.sql` and a test login for each role.
Week 2 (Oct 15 to 21)	Fine assessment, fines and notifications pages, the 3 reports, the overdue-fine trigger. Pair with Austin on checkout and with Farwa on the hold-ready trigger.
Week 3 (Oct 22 to 26)	Merge Oct 22, switch the web app to the Basic plan Oct 25, reset the hosted data, send the email Oct 26.

What you build

You build the money and messages side of the library and the 3 reports, and you also own hosting, the skeleton and the final email.

Tables: Fine, Notification, Loan, Member, plus reads across the whole database for reports.

Build, in this order:

1. **Fine assessment.** When a late return is recorded, add a fine: days late times $0.35, capped at $10 per item. A damaged or lost copy is fined its replacement cost. A loan gets at most one fine of each type.
2. **Fines pages.** A member sees their fines and can pay one (`Is_paid` true with a `Date_paid`). Staff can waive or adjust an amount.
3. **Notifications page.** Lists a member's messages, with a way to mark one as read.
4. **Three reports,** each a formatted, staff-only page built from queries: overdue loans, fines owed per member, and most borrowed items.

Triggers: an overdue fine is created when a late return is recorded, and you pair with Farwa on the hold-ready notification. At least 2 triggers are required across the group.

Also yours: the Azure subscription, MySQL server and web app; the GitHub repo and skeleton; `sample_data.sql` with a test login for each role; and the final email to the TAs.

Watch out for: a fine's paid flag and paid date must agree, and the paid date cannot be before the assessed date; reports need enough sample data to look real.

Done when: returning an overdue item creates the right fine, a member can pay it, a notification appears, and all 3 reports show real rows on the hosted site.

Key dates (all teams)

Date	What
Wed Oct 14	Skeleton live on Azure, login working
Wed Oct 21	Every area's pages merged and deployed
Sat Oct 24	Code freeze: bug fixes only
Mon Oct 26	Final test as each role, email sent to the TAs and uramamur@bcm.edu
Tue Oct 27	Checkpoint #3 deadline (safety day)

Working rules

- Stack: Node.js, Express, the mysql2 driver, plain HTML/CSS/JavaScript, MySQL.
- Test on your own local MySQL copy of the project dump, not on the hosted database.
- Keep database passwords out of GitHub. They go in a local `.env` file that is never committed.
- Merge into main at least every two days. One person owns each file.
- Deactivate or withdraw records instead of deleting them. History tables keep their rows.
- Stuck for more than an hour? Say so in the group chat.